



UNIVERSITI SAINS MALAYSIA

School of Electrical and Electronics Engineering
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EEM 441

Control & Instrumentation Laboratory

Experiment 9

PROJECT CREATION IN SCADA (IOT)

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

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Experiment 9

Project Creation in SCADA (IoT)

Objectives

1. To understand the role and capabilities of SCADA software in an industrial IoT system.
2. To create a working SCADA project, including a Project Node and a SCADA Node.
3. To configure a communication link (Comport) between the SCADA node and an automation device.
4. To create and configure IO Tags and Internal Tags for real-time data acquisition.
5. To configure basic alarms on analog and discrete tags.

Equipment / Software Required

- PC/laptop with Advantech WebAccess/SCADA installed (Project Node and SCADA Node)
- Siemens S7-1200 PLC (or equivalent), with an existing ladder-logic program and known I/O/DB addressing
- Ethernet cable and a configured local network connection between the PC and the PLC
- TIA Portal (or equivalent Siemens engineering software) for reference to the PLC's tag addresses

Introduction

Supervisory Control and Data Acquisition (SCADA) software sits between the plant floor and the people (or systems) that need to monitor and control it. It collects real-time data from automation devices — PLCs, controllers, direct digital control (DDC) systems, and distributed control systems (DCS) — and presents that data through live displays, trends, alarms, and reports.

Advantech WebAccess/SCADA, used in this experiment, is a 100% web-based SCADA package: projects are developed and accessed through a standard web browser, rather than a dedicated desktop client. This lowers the barrier to remote monitoring and multi-site deployment, and it integrates directly with IoT dashboards and cloud platforms via MQTT.

A WebAccess/SCADA system is built from three components:

- **Project Node** — the development platform and web server. All project configuration, graphics, and database files are stored here, and every client connects to it first.
- **SCADA Node** — runs the real-time Kernel (`datacore.exe`), which uses communication drivers to talk to automation hardware (PLCs, I/O modules, controllers) and supplies live data to clients.
- **ViewDAQ Client** — the browser-based interface used to monitor and control the SCADA node: viewing live graphics, historical trends, and alarms, and adjusting set points.

In a small setup (as in this lab), the Project Node and SCADA Node commonly run on the same machine.

Data inside WebAccess/SCADA is represented as **Tags**. There are two broad categories:

- **IO Tags** — real-time measurements and outputs read from, or written to, an automation device. IO Tags can be Analog (floating-point), Discrete (an integer 0–7, commonly used as a 0/1 digital value), or Text (an ASCII string up to 72 characters).
- **Internal Tags** — values that exist inside the SCADA software itself rather than being read from a device: **Constant** tags (a user-defined fixed or operator-editable value), **Calculation** tags (a mathematical/logical expression over up to 20 other tags), and **Accumulation** tags (an integration/totaliser over another tag, e.g. totalising flow into volume).

Both tag categories share the same alarm, security, logging, and reporting features once created.

Table 9.1: Example Siemens S7-1200 address mapping to WebAccess tag addressing.

Siemens S7-1200 address	Data type	Length (bits)	WebAccess address
%I0.4	Bool	1	IX000 (start bit 4)
%Q0.3	Bool	1	QX000 (start bit 3)
%M0.4	Bool	1	MX000 (start bit 4)
%MW0	UInt	16	MW000 (start bit 0)
%IW6.4	UInt	16	IW006 (start bit 4)
%DB1.DBX2.0	Bool	1	DBX1.2 (start bit 0)
%DB3.DBW10	UInt	16	DBW3,10
%DB3.DBD14	Real	64	DBD3,14

Note: for tags addressed inside a Data Block (DB), the offset must be enabled in the block's properties in TIA Portal, since WebAccess addresses DB tags by block number plus offset.

Procedure

Activity 1 — Create a new SCADA project

Objective: create a Project Node and SCADA Node in WebAccess/SCADA.

1. Open a web browser and navigate to 127.0.0.1 (or localhost) to reach the WebAccess/SCADA landing page.
2. Create a new project: enter a **Project Name** and **Project Description**, and set the **Project Node IP Address** (use 127.0.0.1 for a local machine, or the PC's host-name).
3. Leave the **Primary TCP Port** (default 4592) and **HTTP Port** (default 80) at their defaults unless your network/firewall requires otherwise, then click **Submit**.
4. Select the Project Node in the tree structure, click **+**, and choose **New Node** to create the SCADA Node. Fill in the Primary Node Name, Primary/Secondary TCP Port, Node Timeout, and SCADA Node IP Address.
5. Start the Kernel on the SCADA Node (**Start Kernel** from the SCADA Node menu) and confirm it is running by hovering over the WebAccess/SCADA icon, which should report the running project and node name.

Activity 2 — Configure a communication port and add a device

Objective: create the physical/virtual link (Comport) between the SCADA node and the PLC, and add the PLC as a device on that port.

1. Under the SCADA Node, add a new **Comport**. Select the interface type appropriate to your connection:
 - **Serial** — an RS232/RS422/RS485 port (COM1, COM2, ...) on the SCADA node PC.
 - **TCP/IP** — the most common choice for PLCs and IoT devices; WebAccess/SCADA searches all TCP/IP network cards regardless of the assigned Comport number.
 - **OPC** — via an OPC Server, for virtual/software-based links.
2. Configure the port: Port Number, Scan Time (e.g. 1000 ms), and Auto-recover Time.
3. Under the configured Comport, add a new **Device**, selecting the correct Device Type/Driver for your hardware (e.g. Siemens S7-1200, Modbus TCP, Allen-Bradley).
4. Enter the device's communication parameters (IP address, station ID, or baud rate for serial links) and save.

Activity 3 — Create IO Tags

Objective: create IO Tags that map to real signals in the PLC program.

1. Identify at least four signals in your PLC program to bring into SCADA — for example, a Start pushbutton, a Stop pushbutton, an Emergency-stop input, and one analog or word-type value (e.g. a counter or measured value).
2. For each signal, determine its native PLC address and convert it to the equivalent WebAccess address using Table 9.1 as a reference (e.g. %DB1.DBX0.1 for a Start pushbutton, %DB1.DBX0.3 for Stop, %DB1.DBX0.0 for E-stop).

3. Under the Device, click the tag icon and **Insert** a new tag for each signal. Fill in the Tag Type, Tag Name, Address, Start Bit, Conversion Code, and Length fields to match your PLC addressing.
4. Save your tags, then verify each one updates correctly by forcing the corresponding input/output in the PLC and observing the tag's live value in WebAccess.

Activity 4 — Create Internal Tags

Objective: create at least one Constant Tag and one Calculation Tag.

1. Under **Const Point** on the SCADA Node, insert a new Constant Tag (analog type, ConAna, or discrete type, ConDis) — for example, a user-adjustable setpoint value.
2. Under **Calc Point** on the SCADA Node, insert a new Calculation Tag. Choose an analog or discrete result type, then write a mathematical/logical expression referencing one or more of your IO Tags (tags are referenced in the formula by their single-letter designations; the expression is limited to 80 characters).
3. Save and verify that the Calculation Tag's value updates correctly as its input tag(s) change.

Activity 5 — Configure a basic alarm

Objective: enable alarming on at least one analog tag and one discrete tag.

1. Open the configuration for one of your Analog IO Tags and enable a non-zero alarm priority. Analog tags support High-High, High, Low, Low-Low, Deviation, and Rate-of-Change alarms — choose whichever is appropriate to the signal you selected.
2. Open the configuration for one of your Discrete IO Tags and enable state alarming (e.g. alarm when the E-stop input becomes active).
3. Trigger each alarm condition on the PLC side and confirm that the alarm appears correctly in WebAccess.

Report

1. List the Project Node and SCADA Node settings you used (IP address, ports, node names), and explain the role each plays in the system.
2. Present your Comport and Device configuration, including which interface type (Serial/TCP-IP/OPC) you selected and why it was appropriate for your hardware.
3. Tabulate the IO Tags you created: PLC address, WebAccess address, tag type, and a description of what each represents.
4. Present your Constant Tag and Calculation Tag configurations, including the expression used for the Calculation Tag and what it computes.
5. Describe the alarm(s) you configured and demonstrate (with a screenshot or description of your test) that each triggers correctly.

6. Discuss one advantage of WebAccess/SCADA's 100% web-based architecture compared to a traditional desktop-client SCADA system.

References

- MTA SkillLab Sdn Bhd, *i-Factory Training Manual – IIoT System Development Level 1*, Module 3: Project Creation in SCADA.
- Advantech WebAccess/SCADA product documentation: <https://www.advantech.com/en/support/details/installation?id=1-MS9MJV>

